Physiology of the Nervous system

Lecture (2)

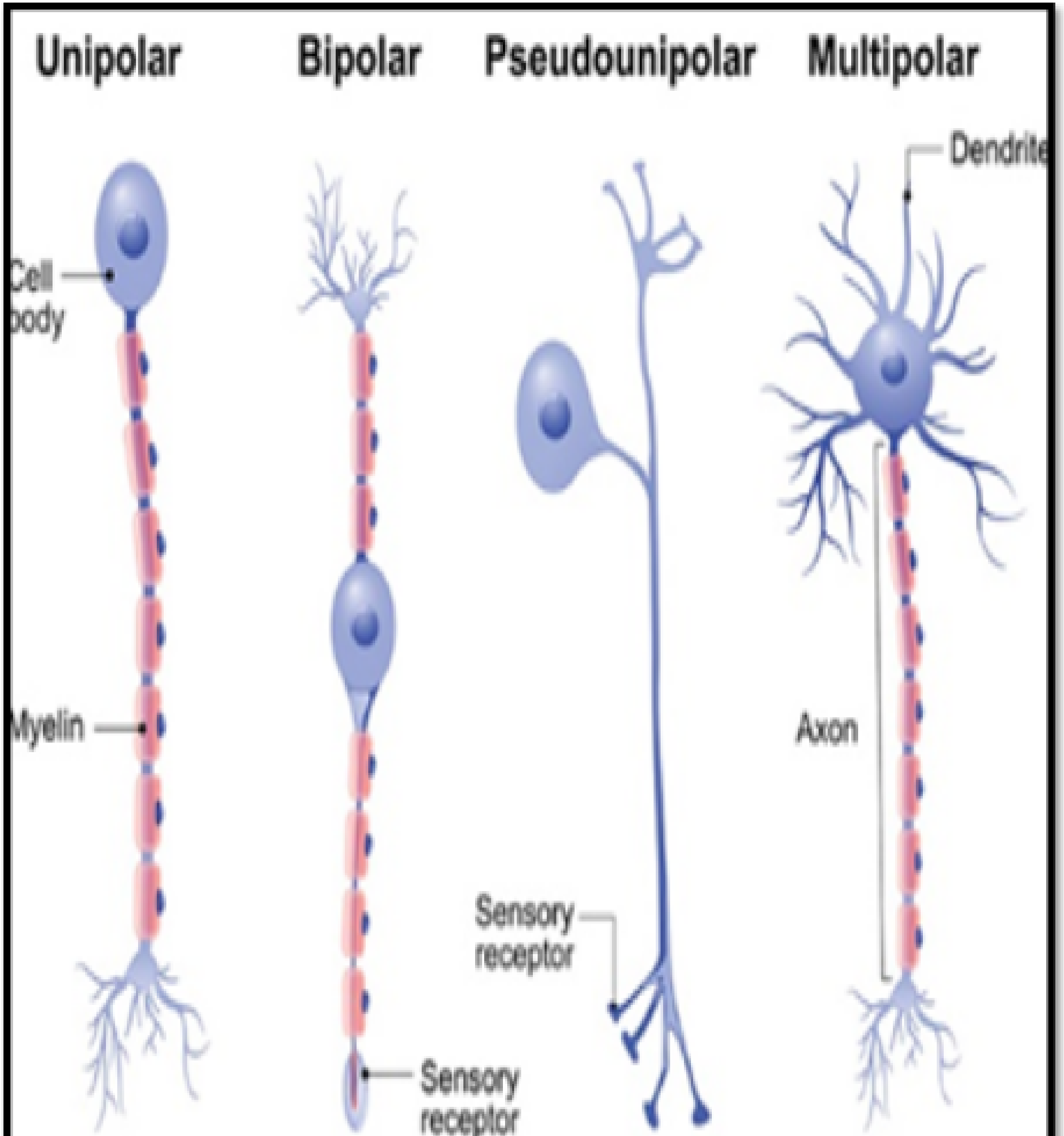
Functional anatomy of the nervous system

Prepared and presented by: Dr. Mazin Salaheldin Abdalla
MBBS
MD (cardiology)
M.Sc. Human physiology
M.Sc. Molecular medicine

Illustrations from (Lippincott illustrated medical physiology)

Introduction

- The anatomical-functional unit of the nervous system is called a **neuron**. A cell that comes in many different shapes and forms, they are, however have the same general structure.
- The function of the neuron is to transmit information (in the form of action potential) across the nervous system as fast as possible
- A neuron is shown next



A neuron

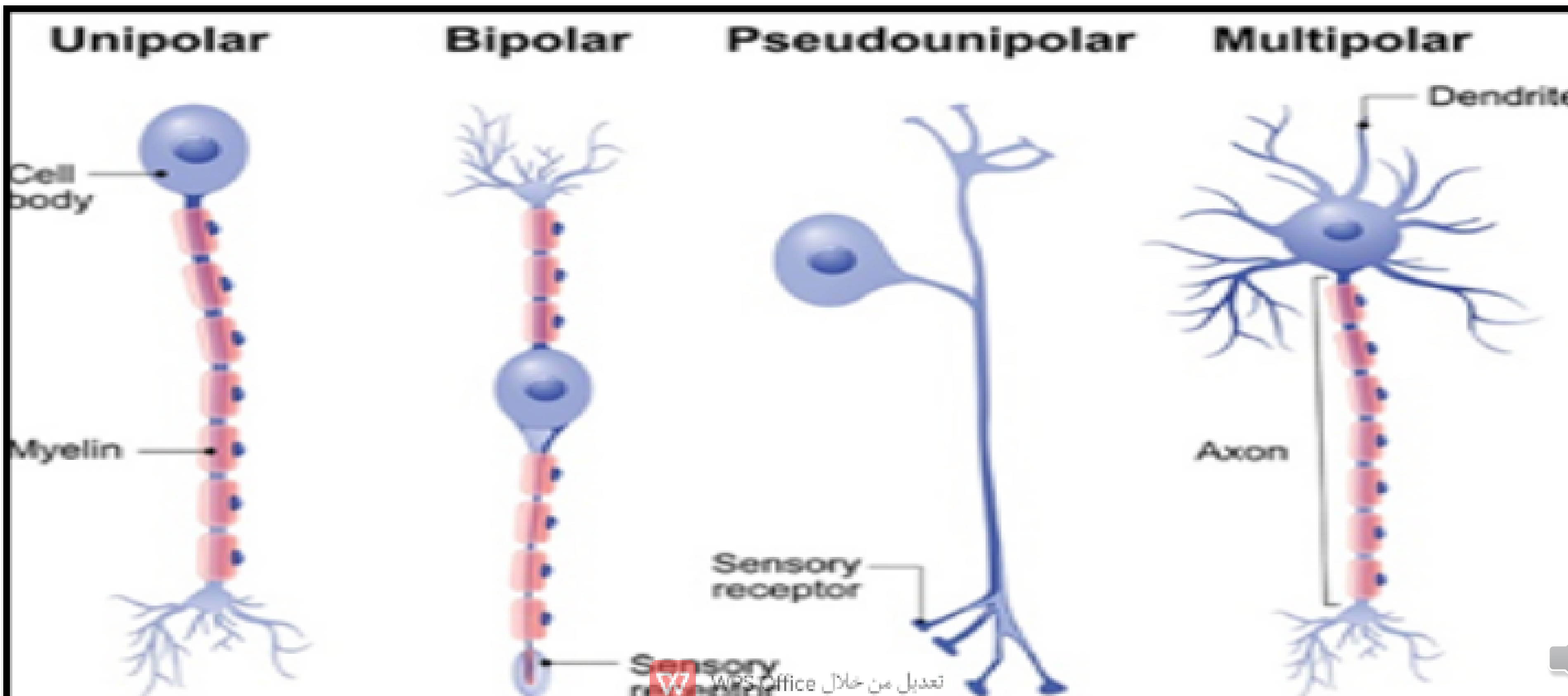
Cell body

Axon: Sending AP away form the cell body

Dendrite(s): Constitute the receptive are and take the AP towards the cell body

Types of Neurons…

- Pseudo-unipolar (sensory)
- Bipolar (most likely specialized sensory)
- Multipolar (most of the CNS)



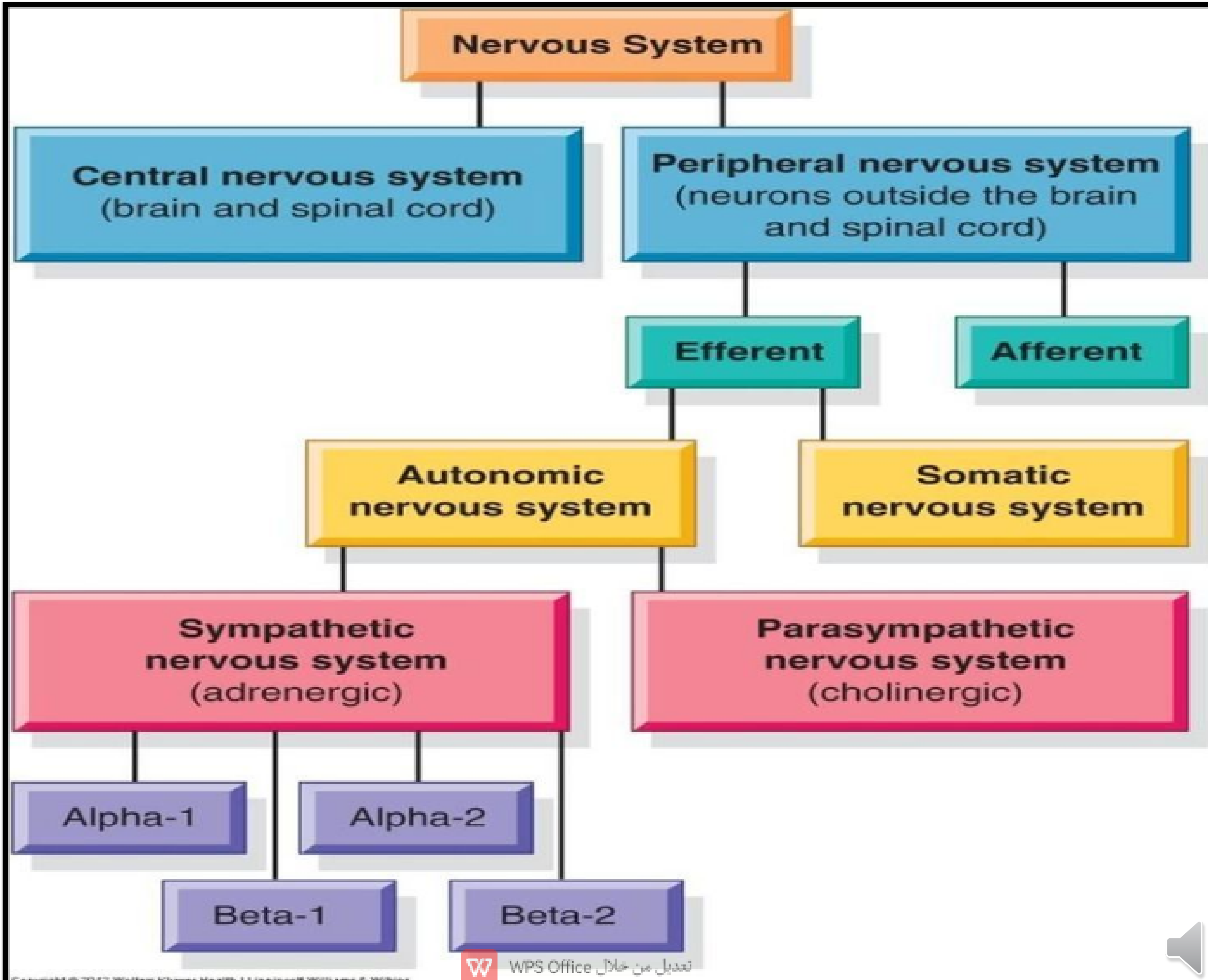
Nervous system organization…

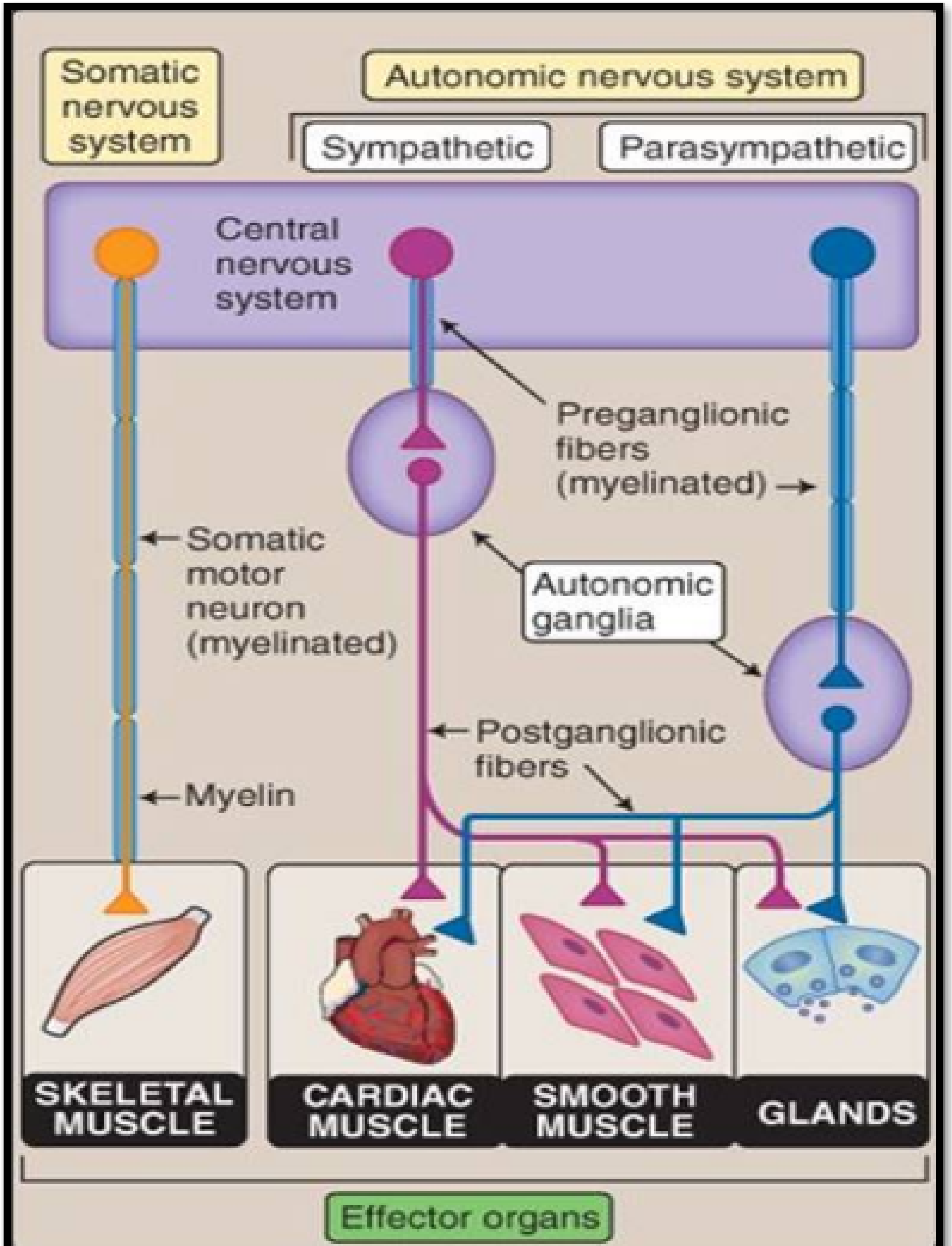
❖ It is divided into: Central nervous system **(CNS)** and peripheral nervous system**(PNS)**

CNS: is composed of Brain and spinal cord

PNS: Everything else

❖ It is divided into: Somatic nervous system **(SNS)** and autonomic nervous system **(ANS)**


Nervous System
Central nervous system (brain and spinal cord)
Peripheral nervous system (neurons outside the brain and spinal cord)
Efferent
Afferent
Autonomic nervous system
Somatic nervous system
Sympathetic nervous system (adrenergic)
Parasympathetic nervous system (cholinergic)
Alpha-1
Alpha-2
Beta-1
Beta-2



THE ANS vs SNS

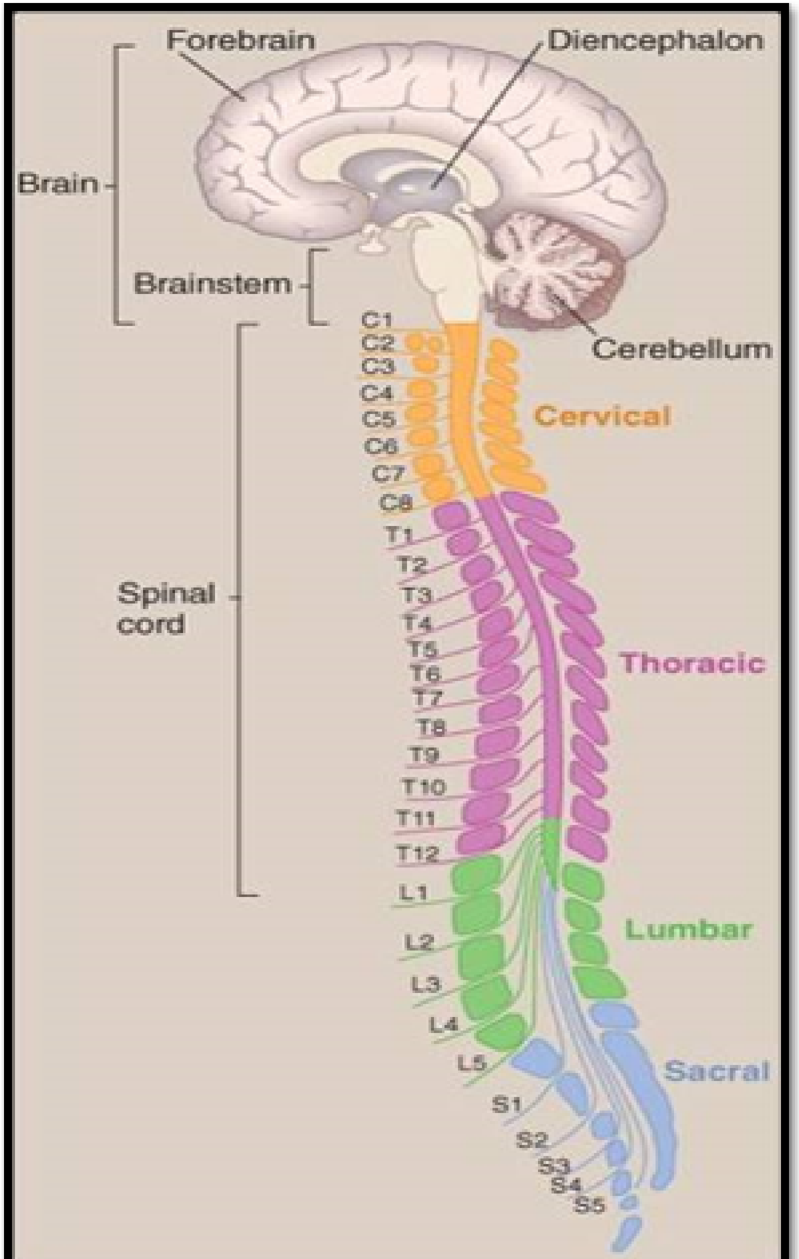
The autonomic vs the somatic nervous system

The ANS controls the smooth muscles while the SNS controls the skeletal muscles

The ANS has two neurons as oppose to the SNS with one neuron

The ANS has acetylcholine and noradrenaline as neurotransmitter and the SNS releases only acetylcholine

CNS (brain and spinal cord)



Both constitute the central nervous system. The spinal cord extends from C1 to L2 and it is divided into segments (cervical to coccygeal)

31 pairs of spinal nerves:

8 cervical

Thoracic 12

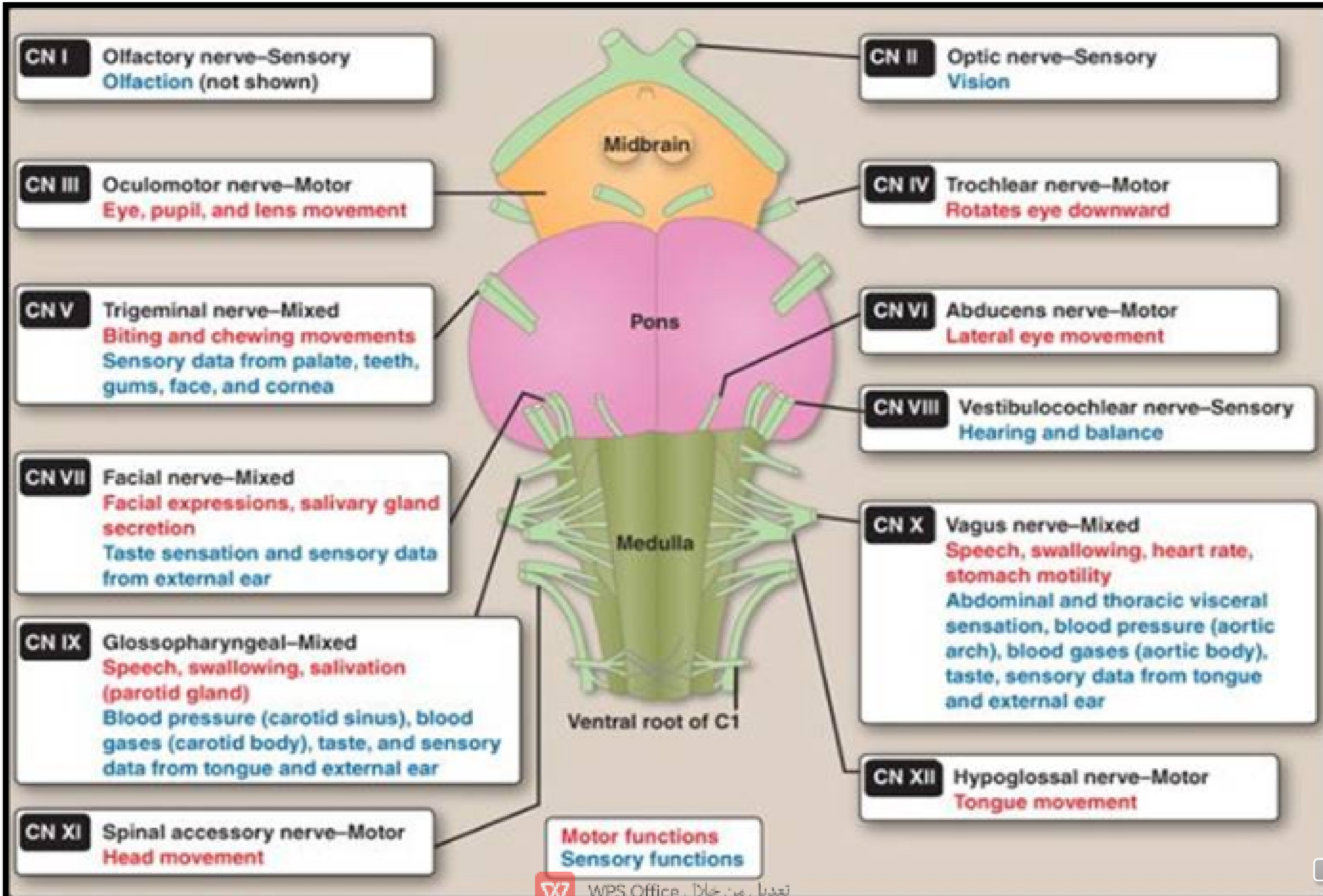
Lumbar 5

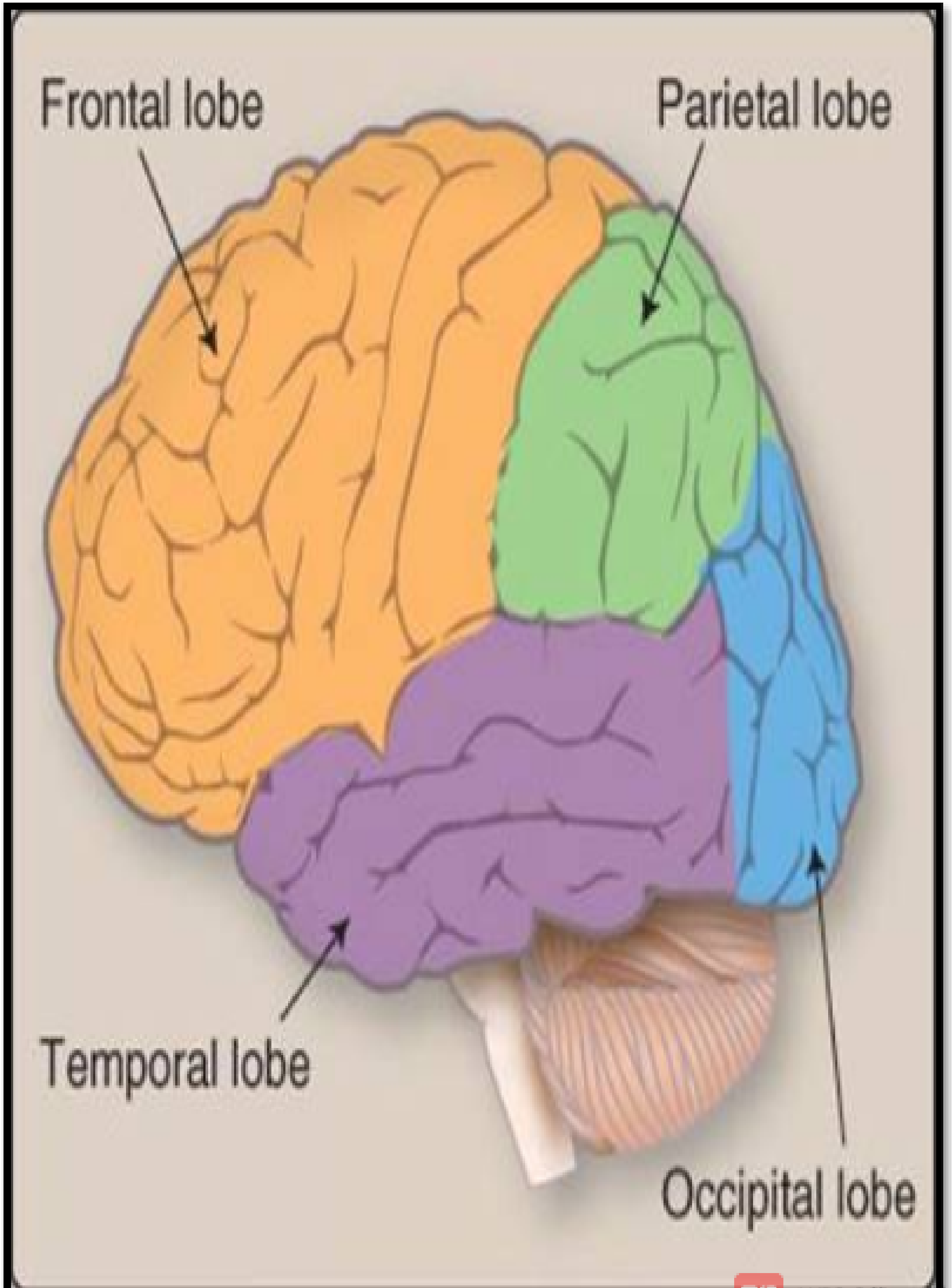
Sacral 5

Coccygeal 1

The spinal cord ends in **Conus medullaris**

The cranial nerves





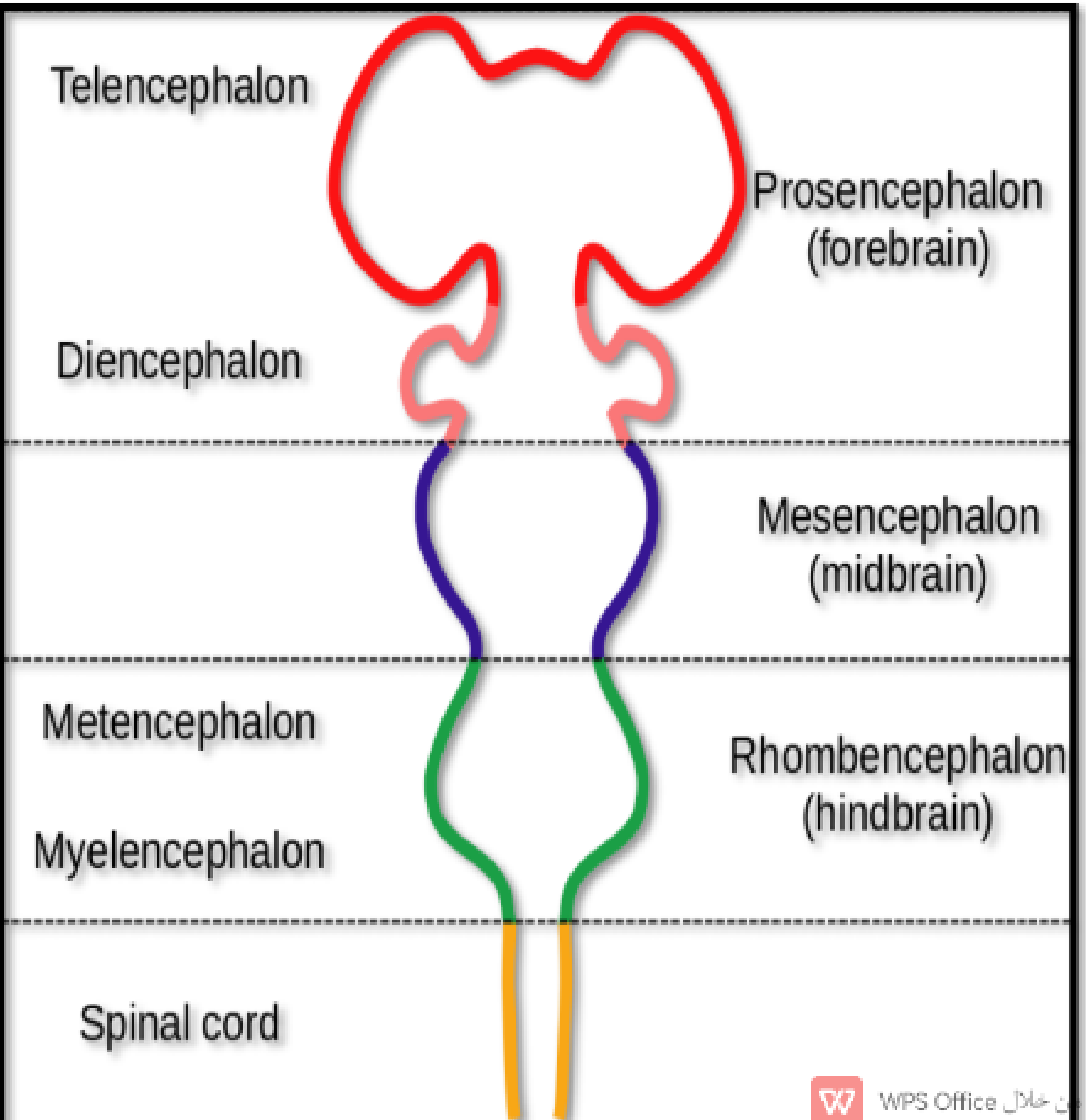
Frontal lobe: Executive function, planning, thinking, organizing, problem solving, emotional behavioural control, personality, social skills

Parietal lobe: Perception, arithmetic, spatial judgement

Temporal lobe: Memory, auditory processing, language understanding

Occipital lobe: Vision

CNS (brain and spinal cord)



During the development of the brain, the prosencephalon produces the telencephalon (**the hemispheres**) and the diencephalon (**thalamus and hypothalamus**)

The mesencephalon stays and develops as the **midbrain**

The rombencephalon produces the **pons** and the **medulla oblongata**

Neuroglia...

These are homeostatic and defensive cells. They are heterogeneous and of different origins. They support the functioning neurons. These cells constitute almost half of the brain. Their presence is paramount to the function of the CNS. Their distribution is in the central, as well in the peripheral system

Oligodendrocytes, astrocytes, ependymal cells and microglia are found in the CNS, while Schwan and satellite cells are in the peripheral system (PNS)

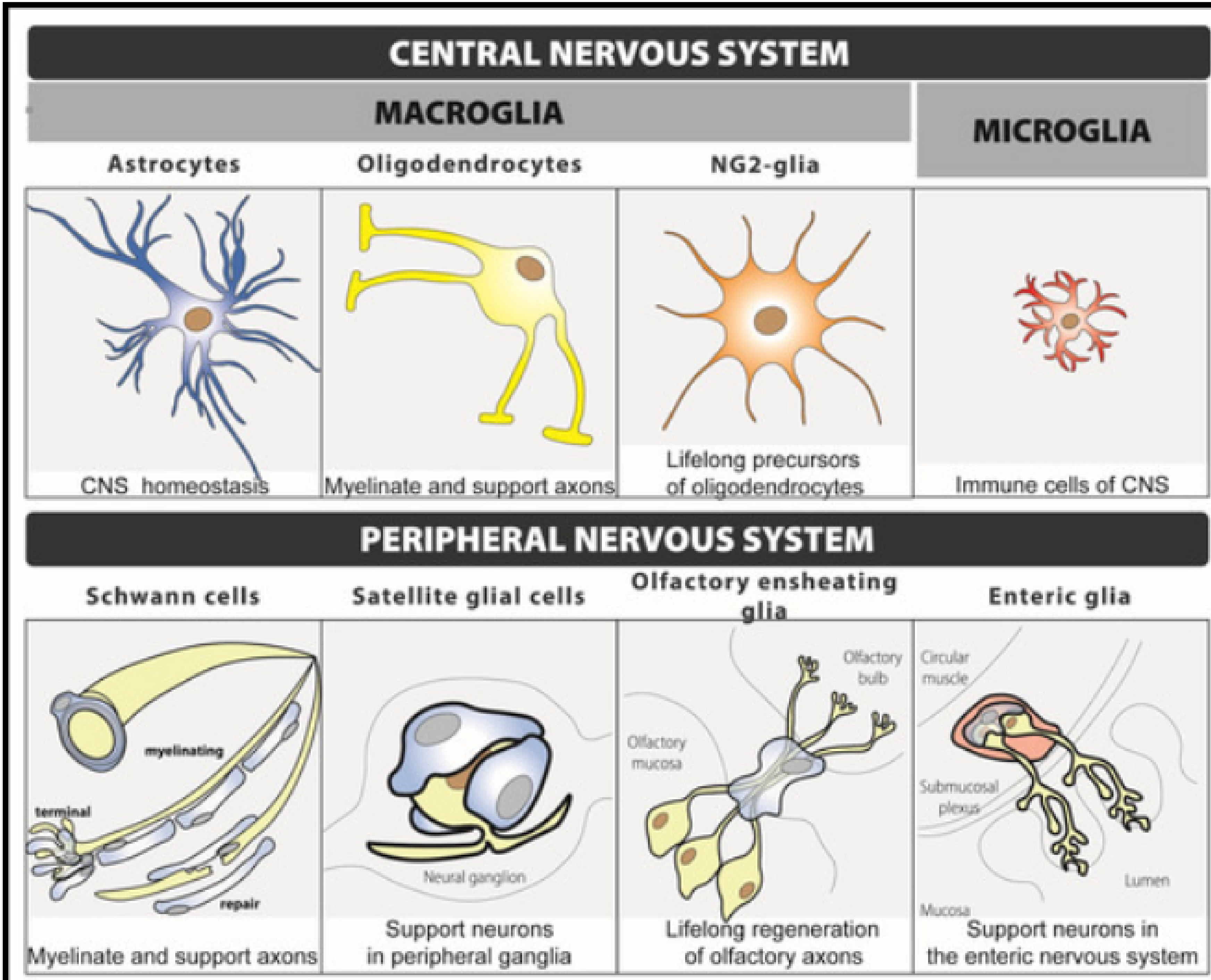
Glial functions…

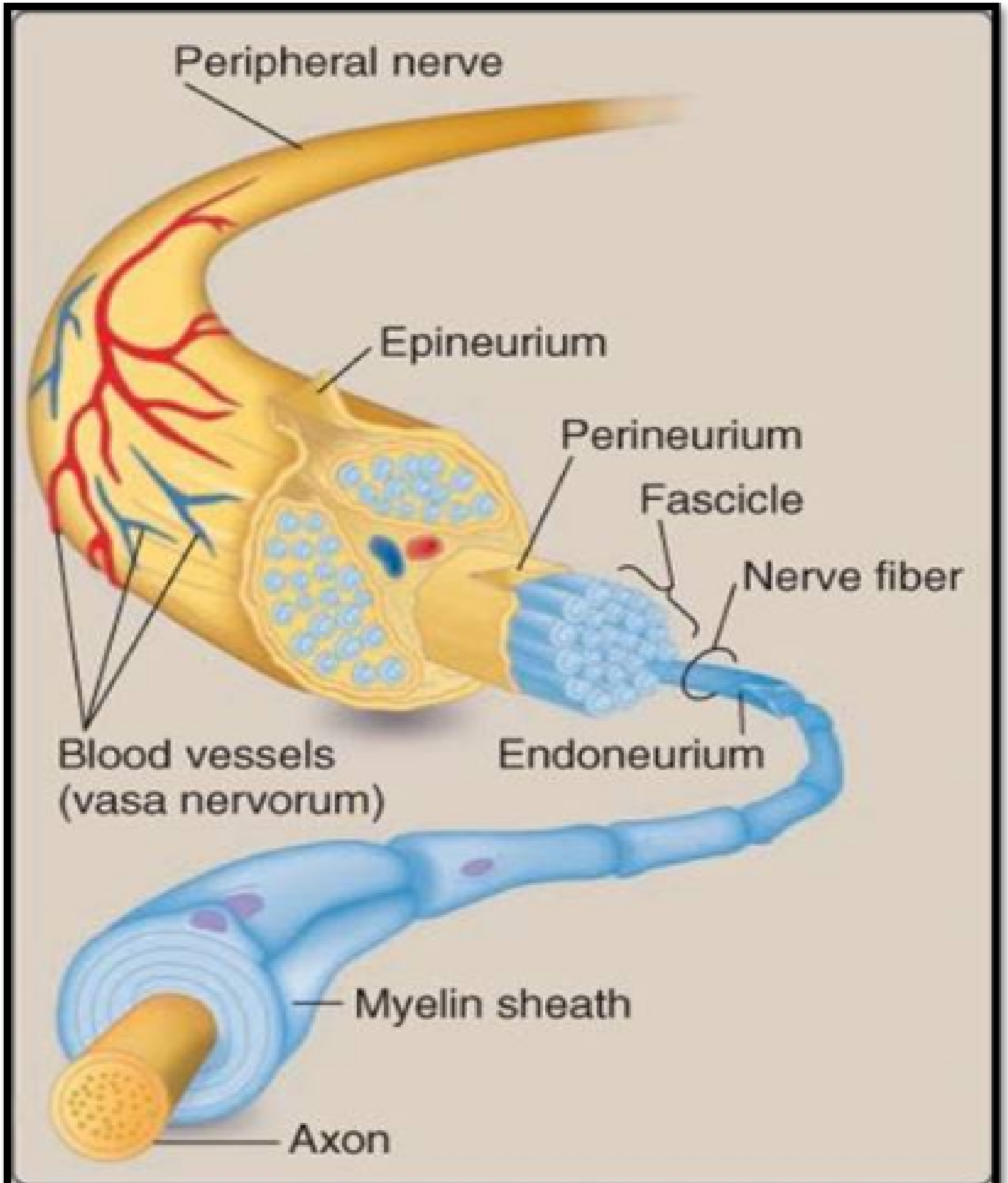
- Oligodendrocytes
- Astrocytes
- Microglia

- Schwan cells
- Satellite cells

- Myelination in the CNS
- Homeostasis
- The immune cell

- Myelination in the PNS
- Support in the ganglia


CENTRAL NERVOUS SYSTEM
MACROGLIA
MICROGLIA
Astrocytes
Oligodendrocytes
NG2-glia
CNS homeostasis
Myelinate and support axons
Lifelong precursors of oligodendrocytes
Immune cells of CNS
PERIPHERAL NERVOUS SYSTEM
Schwann cells
Satellite glial cells
Olfactory ensheating glia
Enteric glia
myelinating
terminal
repair
Neural ganglion
Olfactory bulb
Olfactory mucosa
Circular muscle
Submucosal plexus
Lumen
Mucosa
Myelinate and support axons
Support neurons in peripheral ganglia
Lifelong regeneration of olfactory axons
Support neurons in the enteric nervous system



Nerves belong to the PNS. There are 12 pairs of cranial and 31 pairs of spinal nerves. Nerves could be sensory, motor or mixed. Speed of transmission

A. Thick myelinated
B. Thin myelinated
C. Unmyelinated

Neurotransmitters…

These are chemicals that are released from the presynaptic membrane and cause an action potential in the postsynaptic membrane. Neurotransmitters could be excitatory or inhibitory.

Neurotransmitters…

❖ They could be

Small molecules:

- Glutamate, Glycine and GABA
- Cholinergic: Acetylcholine
- Adrenergic: Adrenaline, Noradrenaline and Dopamine
- Monoamine: Histamine and Serotonin

Neurotransmitters…

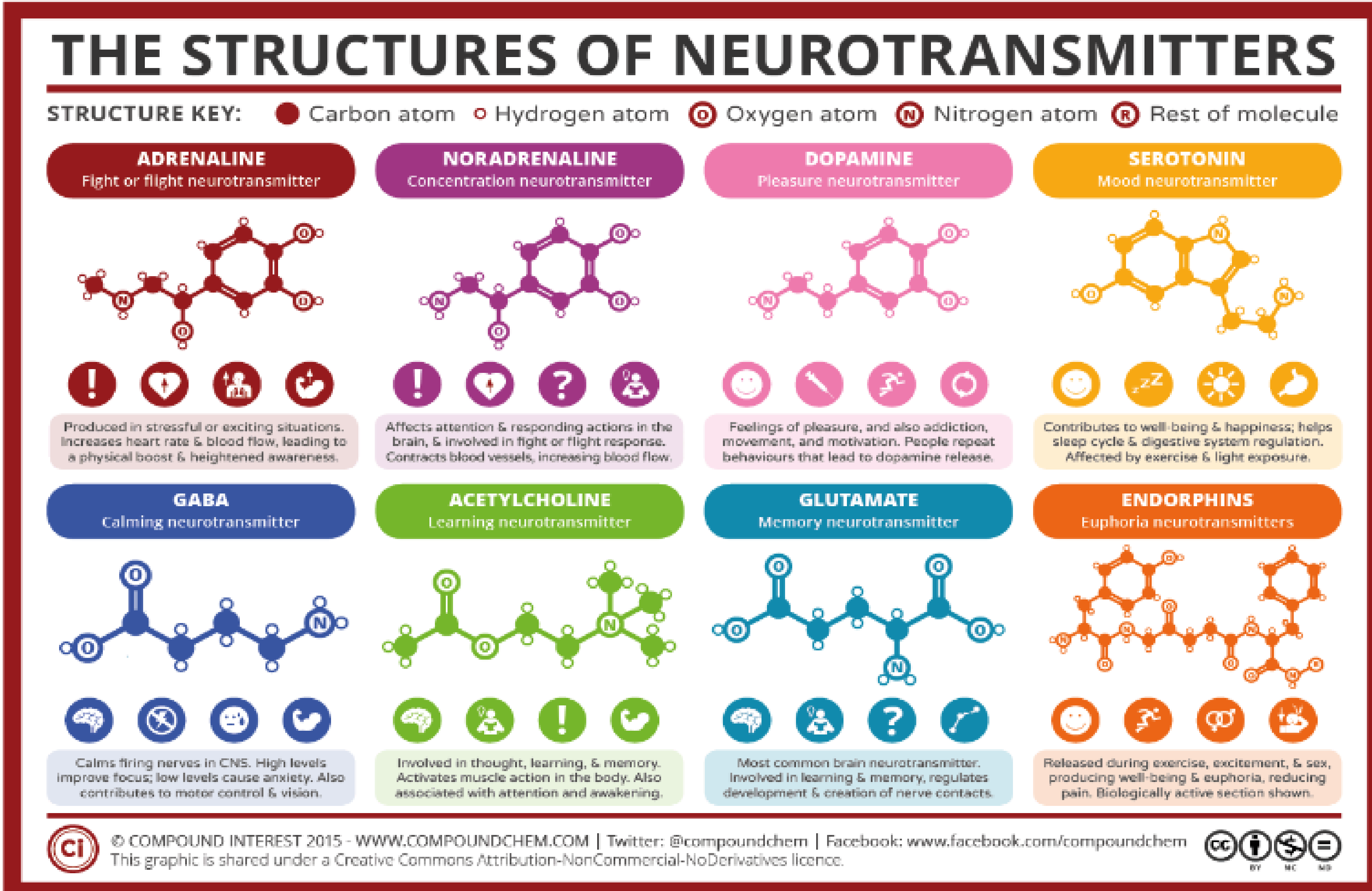
❖ They could be:

- **Peptides:**

Opioids: Dynorphins, Endorphins and Enkephalins

Tachykinin: Neurokinins and Substance P

- **Gases:** NO
- **Purines:** ATP


THE STRUCTURES OF NEUROTRANSMITTERS
STRUCTURE KEY: Carbon atom Hydrogen atom Oxygen atom Nitrogen atom Rest of molecule
ADRENALINE
Fight or flight neurotransmitter
Produced in stressful or exciting situations. Increases heart rate & blood flow, leading to a physical boost & heightened awareness.
NORADRENALINE
Concentration neurotransmitter
Affects attention & responding actions in the brain, & involved in fight or flight response. Contracts blood vessels, increasing blood flow.
DOPAMINE
Pleasure neurotransmitter
Feelings of pleasure, and also addiction, movement, and motivation. People repeat behaviours that lead to dopamine release.
SEROTONIN
Mood neurotransmitter
Contributes to well-being & happiness; helps sleep cycle & digestive system regulation. Affected by exercise & light exposure.
GABA
Calming neurotransmitter
Calms firing nerves in CNS. High levels improve focus; low levels cause anxiety. Also contributes to motor control & vision.
ACETYLCHOLINE
Learning neurotransmitter
Involved in thought, learning, & memory. Activates muscle action in the body. Also associated with attention and awakening.
GLUTAMATE
Memory neurotransmitter
Most common brain neurotransmitter. Involved in learning & memory, regulates development & creation of nerve contacts.
ENDORPHINS
Euphoria neurotransmitters
Released during exercise, excitement, & sex, producing well-being & euphoria, reducing pain. Biologically active section shown.
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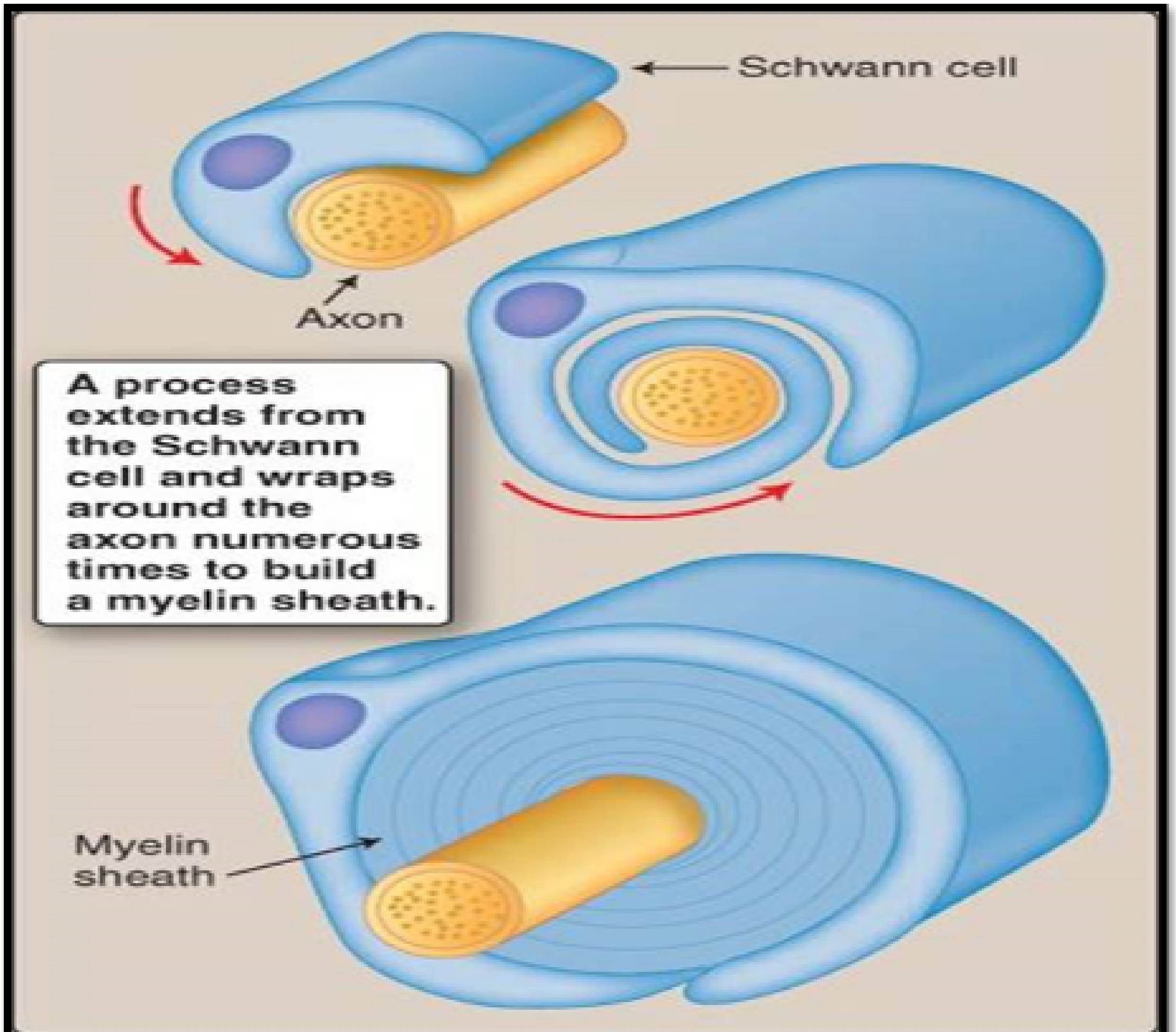
Myelination… (an insulator)

- Lipid rich material, covers the axon and sometimes dendrites.
- The wrapping around the axon is called myelin sheath
- It increases the speed of conduction in the axon by the mode of **salutatory effect** (jumping) from 10m/s to 150m/s
- Also decreases the loss of electrical signal to the surroundings

Myelination… (an insulator)



The myelin sheath wraps around the axon. Produced by the Schwan cell. The wrapping leaves the nodes of **Ranvier** open for the propagation of the action potential (salutatory effect) hence speeding the transmission

Some demyelination diseases…

- Multiple sclerosis (MS)
- Central pontine myelnolysis (CPM)
- Guillian Barre

Physiology lit. (recommended)…

- Guyton and Hall textbook of physiology
- Lippincott illustrated physiology
- Ganong review of medical physiology
- Linda Costanzo textbook of medical physiology
- BRS, Linda Costanzo
- Essential to physiology